

Evidence-bound verification for Uniswap v3 replays

Satoshi Kawasaki | Research & Implementation | Grant application proposal

Proposed request: \$46,000 | 500 person-hours | 28 weeks at 20 h/week

The review problem

Uniswap v3 tooling maintainers need to trace a replay comparison to the inputs, code and checks behind it. Matching state values does not establish complete event history; a rejected test may fail for an unrelated reason. This project makes those limits inspectable.

Inspect the existing demo

Four public synthetic cases exercise selected pool/tick-state reconstruction, declared missing coverage and malformed input. Python 3.12.6 and the standard library; no RPC endpoint or wallet. This completed baseline is not billed to the grant. It does not discover secretly omitted logs or provide OS isolation.

[Open the fixed public demo and run instructions](#)

The remaining work

C1 has a v3 replay baseline with partial evidence integration. Expected-result protection across the full declared workflow, event-specific explanation checks and execution evidence for seven original claims remain incomplete. Those claims currently remain OPEN.

Funded deliverables and acceptance evidence

Deliverable	Plan	Acceptance evidence
M1 Protected comparison execution	Weeks 1-6 \$10,480	Allowed/denied controls across every registered execution target; separate expected results and outer grading.
M2 Reason and event attribution controls	Weeks 7-12 \$9,560	Baseline, reached check and correct reason/event; original semantic mutants plus rejection controls.
M3 Execution-bound claim review	Weeks 13-23 \$17,840	Seven claim reviews, each bound to retrievable same-scope evidence and unresolved dependencies.
M4 Reproducible reference release	Weeks 24-28 \$8,120	Clean-environment synthetic reproduction; source, tests, dependency/licence records and report.

Acceptance and delivery

All registered observable checks must be implemented and executed. Every claim need not become supported; missing evidence and direct refutations remain visible. Public synthetic reproduction is separate from restricted empirical evidence. Source, tests, environment and licence records, and a final report are included.

Scope and accountability

Finite v3 research and software delivery; v4 hooks, trading and hosted operations are outside scope. Adoption and review-time savings are unmeasured. Generative AI substantially assists implementation, writing and review; Satoshi Kawasaki is responsible for the claims. AI review is not independent human validation.

Budget basis: 500 person-hours at \$90 + \$1,000 incremental tool/compute allowances. Progress reporting is included. Budget and schedule remain provisional until the final worker/resource matrix and experiment contexts are registered. Allocation, start, licence review and acceptance-remediation terms require agreement. This is a planning estimate, not an optimal-price claim.